Q1a) Compare doGet() and doPost() methods in Servlet

Basis	doGet()	doPost()
Purpose	Used to request data from server	Used to send data to server
Data Visibility	Data is visible in URL	Data is hidden from URL
Security	Less secure	More secure
Data Length	Limited length of data	No major size limitation
Speed	Faster	Slightly slower
Bookmarking	Possible	Not possible
Browser History	Data stored in history	Data not stored in history
Use Case	Searching, fetching records	Login forms, registration forms
Method Syntax	doGet(HttpServletRequest req, HttpServletResponse res)	doPost(HttpServletRequest req, HttpServletResponse res)

Example:

```
protected void doGet(HttpServletRequest req,
HttpServletResponse res)
throws ServletException, IOException
{
   res.getWriter().println("This is GET method");
}
```

```
protected void doPost(HttpServletRequest req,
HttpServletResponse res)
throws ServletException, IOException
{
   res.getWriter().println("This is POST
method");
}
```

Q1b) What is Servlet? Explain Servlet Life Cycle with Example.

A **Servlet** is a server-side Java program used to create dynamic web applications.
It runs on a web server and handles client requests and responses.

Servlet belongs to the package: javax.servlet

Uses of Servlet - Processing client requests, Generating dynamic web pages, Handling form data, Session management, Database connectivity

Servlet Life Cycle - Servlet life cycle defines how a servlet is loaded, initialized, processes requests, and destroyed by the web container. Servlet life cycle consists of the following stages:

1. Loading and Instantiation
2. Initialization (init())
3. Request Processing (service())
4. Destruction (destroy())

Diagram of Servlet Life Cycle

Explanation of Methods

1) init()

- Called only once
- Used for initialization
- Executed when servlet is loaded

2) service()

- Called for every client request
- Processes request and generates response
- Internally calls doGet() or doPost()

3) destroy()

- Called once before removing servlet
- Used to release resources

Example Program of Servlet Life Cycle

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class LifeCycleServlet extends HttpServlet
{
   public void init()
   {
```

```java
        System.out.println("Servlet Initialized");
    }
    public void service(ServletRequest req, ServletResponse res)
   {
       System.out.println("Request Processed");
   }
    public void destroy()
    {
        System.out.println("Servlet Destroyed");
    }
}
```

Servlet life cycle defines how a servlet is created, initialized, used to process requests, and destroyed by the web container. It helps in efficient resource management and dynamic web application development.

Q2a) What is Session? List Session Tracking Techniques. How Cookies are Used to Track a Session.

A **session** is the time interval during which a user interacts with a web application between login and logout.

HTTP protocol is **stateless**, meaning it does not remember previous requests. Therefore, session tracking techniques are used to maintain user information across multiple requests.

Session Tracking Techniques - The following techniques are used for session tracking:

1. Cookies - Cookies are small text files stored in the client browser.
2. URL Rewriting - In URL rewriting, session information is appended to the URL.
3. Hidden Form Fields - Hidden values are stored inside HTML forms using hidden input fields.
4. HttpSession Object - HttpSession object stores user-specific information at server side.
5. SSL Session Tracking - Secure Socket Layer (SSL) uses encrypted connections for session tracking.

Cookies for Session Tracking

A **cookie** is a small text file stored on the client browser by the server.

It stores user-specific information such as Username, Session ID, Login details, Preferences

When the user sends another request, the browser sends the cookie back to the server, helping identify the user session.

Steps of Cookie-Based Session Tracking

1. User sends request to server
2. Server creates cookie object
3. Cookie is stored in browser
4. Browser sends cookie with every request
5. Server identifies the user using cookie data

Diagram

```
Client Browser              Web Server
     |                            |
     |---- Request --------------->|
     |                            |
     |<--- Cookie Created --------|
     |                            |
Cookie stored in browser          |
     |                            |
     |---- Sends Cookie --------->|
     |                            |
     |<--- Response --------------|
```

Example Program

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class CookieServlet extends HttpServlet
{
   public void doGet(HttpServletRequest req, HttpServletResponse res)
   throws IOException
   {
      res.setContentType("text/html");

      Cookie c = new Cookie("user","Admin");

      res.addCookie(c);

      PrintWriter out = res.getWriter();
```

```
        out.println("Cookie Created");
    }
}
```

Advantages of Cookies - Simple to use, Maintains session information, Improves user experience.

Disadvantages - Less secure, Limited storage size, User can disable cookies.

Session tracking helps maintain user state in web applications. Cookies are one of the most commonly used techniques for tracking sessions between client and server.

Q2b) Distinguish between External and Internal DTDs

Basis	**Internal DTD**	**External DTD**
Definition	DTD declared inside XML document	DTD declared in separate file
Location	Written within XML file	Stored in external .dtd file
Reusability	Cannot be reused easily	Can be reused by many XML files
Maintenance	Difficult for large documents	Easy to maintain
File Size	Makes XML file lengthy	Keeps XML file smaller
Syntax	Uses <!DOCTYPE root [ ]>	Uses SYSTEM or PUBLIC keyword
Usage	Small XML documents	Large applications
Ex	<!DOCTYPE student [<!ELEMENT student (name,roll)> <!ELEMENT name (#PCDATA)> <!ELEMENT roll (#PCDATA)>]>	**XML File** <!DOCTYPE student SYSTEM "student.dtd"> **student.dtd File** <!ELEMENT student (name,roll)> <!ELEMENT name (#PCDATA)> <!ELEMENT roll (#PCDATA)>

a) Explain HttpServlet Class with Example. Explain the Servlet Life Cycle Methods.

The HttpServlet class is a part of Java Servlet API used to create web applications based on the HTTP protocol.

It belongs to package: javax.servlet.http

HttpServlet extends GenericServlet and provides methods to handle HTTP requests such as: GET, POST, PUT, DELETE.

Important Methods of HttpServlet

Method	Purpose
doGet()	Handles GET request
doPost()	Handles POST request
doPut()	Handles PUT request
doDelete()	Handles DELETE request
service()	Dispatches request to appropriate method

Example of HttpServlet

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
public class MyServlet extends HttpServlet
{
   public void doGet(HttpServletRequest req, HttpServletResponse res)
   throws IOException
   {
      res.setContentType("text/html");

      PrintWriter out = res.getWriter();

      out.println("<h2>Hello from HttpServlet</h2>");
   }
}
```

HttpServlet simplifies development of web applications by providing HTTP-specific functionalities. Servlet life cycle methods help manage servlet creation, request processing, and destruction efficiently.

Q1b) Explain XML using DTDs & XML using XML Schema with Example.

1) XML using DTD - DTD (Document Type Definition) defines the structure and legal elements of an XML document.

It specifies: Elements, Attributes, Order of elements

Example of XML using DTD

XML File

```
<?xml version="1.0"?>
<!DOCTYPE student [
<!ELEMENT student (name,roll)>
<!ELEMENT name (#PCDATA)>
<!ELEMENT roll (#PCDATA)>
]>

<student>
   <name>Rahul</name>
   <roll>101</roll>
</student>
```

Advantages of DTD - Simple to write, Defines XML structure, Validates XML document

Limitations - No data type support, Limited validation capability, Not extensible

2) XML using XML Schema - XML Schema (XSD) is used to define the structure and data types of XML documents.

It is more powerful than DTD.

Features: Supports data types, Supports namespaces, Better validation

Example of XML Schema

XML File

```
<?xml version="1.0"?>
<student xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:noNamespaceSchemaLocation="student.xsd">

   <name>Rahul</name>
   <roll>101</roll>

</student>
```

XSD File (student.xsd)

```
<?xml version="1.0"?>

<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="student">
```

```
    <xs:complexType>
      <xs:sequence>
        <xs:element name="name" type="xs:string"/>
        <xs:element name="roll" type="xs:int"/>
      </xs:sequence>
    </xs:complexType>
</xs:element>
</xs:schema>
```

DTD and XML Schema are used to validate XML documents. XML Schema is more powerful and flexible compared to DTD because it supports data types and advanced validation features.

Q2a) Explain Servlet Session Management & Tracking Methods

Session management is used to maintain the state of a user between multiple HTTP requests.

HTTP is a **stateless protocol**, meaning the server does not remember previous requests.
Therefore, session management techniques are used to identify users and store their information during interaction with a web application.

Need of Session Tracking - To maintain login information, Shopping cart management, Online banking applications, User personalization

Session Tracking Methods in Servlet -

1) Cookies - Cookies are small text files stored in the client browser.

They store user information like Username, Session ID, Preferences

Example - Cookie c = new Cookie("user","admin");
response.addCookie(c);

Advantages - Simple to use, Automatic storage in browser

Disadvantages - Less secure, User may disable cookies

2) URL Rewriting - In URL rewriting, session information is appended to the URL.

Example - response.sendRedirect("welcome?user=Mayur");

Advantages - Works even if cookies are disabled

Disadvantages - URL becomes lengthy, Less secure because data is visible in URL

3) Hidden Form Fields - Hidden values are stored inside HTML forms using hidden input fields.

Example - <input type="hidden" name="user" value="Rahul">

Advantages - Easy to implement

Disadvantages - Works only with forms

4) HttpSession API - HttpSession object stores user-specific information at server side.

Example - HttpSession session = request.getSession();
session.setAttribute("user","Rahul");

Advantages - More secure, Server-side storage

Disadvantages - Consumes server memory

5) SSL Session Tracking - Secure Socket Layer (SSL) uses encrypted connections for session tracking.

Advantages - Highly secure

Disadvantages - More processing overhead

Diagram of Session Tracking

```
Client Browser <------> Web Server
      |                  |
      |---- Session ID ---->|
      |<--- Response -------|
```

Servlet session management helps maintain user information across multiple requests. Different tracking methods such as cookies, URL rewriting, hidden fields, and HttpSession are used depending on security and application requirements.

b) Write Note on AJAX

AJAX stands for Asynchronous JavaScript And XML

AJAX is a web development technique used to create fast and interactive web applications without reloading the entire webpage.

It allows data exchange between client and server asynchronously in the background.

Features of AJAX - Asynchronous communication, Faster response, Partial page update, Improved user experience, Reduces server load

Working of AJAX

1. User sends request
2. JavaScript creates XMLHttpRequest object

3. Request sent to server
4. Server processes request
5. Response returned asynchronously
6. Webpage updated without reload

AJAX Architecture Diagram -

```
User
 |
 v
JavaScript
(XMLHttpRequest)
 |
 v
Web Server
 |
 v
Database
```

Example of AJAX

```
<html>
<head>
<script>
function loadData()
{
   var xhttp = new XMLHttpRequest();

   xhttp.onreadystatechange = function()
   {
      if(this.readyState == 4 && this.status == 200)
      {
         document.getElementById("demo").innerHTML = this.responseText;
      }
   };
   xhttp.open("GET","data.txt",true);
   xhttp.send();
}
</script>
</head>
<body>
<button onclick="loadData()">
  Get Data
</button>
<div id="demo"></div>
</body>
</html>
```

Advantages of AJAX - Faster web applications, No full page reload, Better user interaction, Reduced bandwidth usage

Disadvantages of AJAX - JavaScript must be enabled, Debugging is difficult, Browser compatibility issues

Applications of AJAX - Google Maps, Gmail, Live search, Chat applications

AJAX improves web application performance by enabling asynchronous communication between client and server, resulting in faster and more interactive user experiences.

Q1b) What is XML DTDs? Explain with example. Differentiate XML DTDs Vs XML schema (Min.04).

Basis	**XML DTD**	**XML Schema (XSD)**
Full Form	Document Type Definition	XML Schema Definition
Syntax	Non-XML syntax	XML-based syntax
Data Types	Does not support data types	Supports data types
Validation	Limited validation	Strong validation
Namespace Support	Not supported	Supported
Extensibility	Less extensible	Highly extensible
Complexity	Simple	More powerful and flexible
Reusability	Less reusable	More reusable
Data Restrictions	Cannot define restrictions	Can define restrictions like min/max
Support for Objects	No support	Supports complex/simple types

Q2a) Explain the Servlet and MySQL Database Connectivity with Example Code to Display Data from Employee Table

Servlet and MySQL Connectivity

Servlet can interact with databases using **JDBC (Java Database Connectivity)**.

JDBC is an API used to connect Java applications with databases like MySQL.

Steps for Servlet-MySQL Connectivity

1. Import JDBC packages
2. Load JDBC Driver
3. Establish connection
4. Create statement
5. Execute SQL query
6. Process result

7. Close connection

Employee Table

```
CREATE TABLE employee
(
   emp_id INT,
   emp_name VARCHAR(20),
   emp_dept VARCHAR(20)
);
```

Example Program

```
import java.io.*;
import java.sql.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class EmployeeServlet extends HttpServlet
{
   public void doGet(HttpServletRequest req, HttpServletResponse res)
   throws ServletException, IOException
   {
      res.setContentType("text/html");
      PrintWriter out = res.getWriter();

      try {
         // Load Driver
         Class.forName("com.mysql.jdbc.Driver");

         // Establish Connection
         Connection con = DriverManager.getConnection(
         "jdbc:mysql://localhost:3306/test",
         "root",
         "root");

         // Create Statement
         Statement stmt = con.createStatement();

         // Execute Query
         ResultSet rs = stmt.executeQuery(
         "SELECT * FROM employee");

         out.println("<h2>Employee Records</h2>");
```

```
            out.println("<table border='1'>");
            out.println("<tr>");
            out.println("<th>ID</th>");
            out.println("<th>Name</th>");
            out.println("<th>Department</th>");
            out.println("</tr>");

            while(rs.next())
            {
               out.println("<tr>");

               out.println("<td>" + rs.getInt("emp_id") + "</td>");

               out.println("<td>" + rs.getString("emp_name") + "</td>");

               out.println("<td>" + rs.getString("emp_dept") + "</td>");

               out.println("</tr>");
            }
            out.println("</table>");
            con.close();
         }  catch(Exception e) {
            out.println(e);
         }
      }
   }
```

Output

Employee Records

ID	Name	Department
1	Rahul	IT
2	Amit	HR
3	Neha	Sales

Advantages of JDBC Connectivity - Easy database interaction, Supports multiple databases, Dynamic data retrieval, Used in enterprise applications

Servlet and MySQL connectivity using JDBC helps create dynamic web applications by storing and retrieving data from databases efficiently.

Q2b) Write Note on XML

XML stands for Extensible Markup Language

XML is used to store and transport data in a structured format.

It is designed to be Self-descriptive, Platform independent & Human readable

Features of XML - Extensible language, User-defined tags, Stores structured data, Platform independent, Supports data sharing

Structure of XML Document

```
<?xml version="1.0"?>
<employee>
   <emp_id>101</emp_id>
   <emp_name>Rahul</emp_name>
   <emp_dept>IT</emp_dept>
</employee>
```

Components of XML

1. Declaration
2. Elements
3. Attributes
4. Root Element
5. Tags

XML Declaration Syntax - XML declaration appears at the beginning of XML document. <?xml version="1.0" encoding="UTF-8"?>

Attribute	Meaning
version	Specifies XML version
encoding	Specifies character encoding

XML Namespace - Namespace is used to avoid naming conflicts between XML elements.

Namespaces uniquely identify elements using URI references.

Syntax of Namespace - <book xmlns:bk="http://www.example.com/book">

Example of Namespace -

```
<?xml version="1.0"?>

<bk:book

xmlns:bk="http://www.example.com/book">
```

<bk:title>XML Basics</bk:title>

</bk:book>

Advantages of XML - Easy data exchange, Simple and readable, Supports platform independence, Used in web services.

Disadvantages of XML - Large document size, Slower processing, More complex than HTML

Applications of XML - Web services, Data storage, Configuration files, Data transfer between applications

XML is widely used for storing and exchanging data between different systems because of its flexibility, readability, and platform independence.

Q1a) What are the Strengths of XML Technology? Explain the Need of XML.

XML is used to store, transport, and exchange data between different systems.

Strengths of XML Technology

1) Platform Independent - XML works on all platforms and operating systems.

2) User-Defined Tags - Users can create their own tags according to application requirements.

3) Self Descriptive - XML data is easy to understand because tags describe the data itself.

4) Data Storage and Transport - XML is mainly used to store and transfer structured data over networks.

5) Extensible - New elements and tags can be added easily without affecting existing data.

6) Human and Machine Readable - XML documents can be read by both humans and machines.

7) Supports Unicode - XML supports multiple languages through Unicode.

8) Used in Web Services - XML is widely used in SOAP, APIs, Configuration files

Need of XML - XML is needed because HTML only displays data but does not store or describe data properly.

Reasons for Need of XML

1) Data Exchange - XML allows data exchange between different applications and systems.

2) Data Storage - XML stores structured and organized data.

3) Platform Independence - Applications developed on different platforms can communicate using XML.

4) Simplifies Data Sharing - XML simplifies sharing of information over the internet.

5) Separation of Data and Presentation - XML stores only data, while presentation is handled separately.

Example of XML

```xml
<?xml version="1.0"?>
<employee>
   <emp_id>101</emp_id>
   <emp_name>Rahul</emp_name>
   <department>IT</department>
</employee>
```

XML is a powerful technology used for storing, transporting, and sharing data across different platforms and applications efficiently.

Q2c) Write a Servlet Which Accepts Two Numbers Using POST Method and Displays the Maximum Number

HTML Form

```html
<html>
<body>
<form action="MaxServlet" method="post">
Enter First Number:
<input type="text" name="n1"><br><br>
Enter Second Number:
<input type="text" name="n2"><br><br>
<input type="submit" value="Find Maximum">
</form>
</body>
</html>
```

Servlet Program

```java
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class MaxServlet extends HttpServlet
{
```

```
    public void doPost(HttpServletRequest req, HttpServletResponse res)
    throws ServletException, IOException
    {
        res.setContentType("text/html");
        PrintWriter out = res.getWriter();

        int n1 = Integer.parseInt(
        req.getParameter("n1"));

        int n2 = Integer.parseInt(
        req.getParameter("n2"));

        int max;
        if(n1 > n2)
        {
            max = n1;
        }
        else
        {
            max = n2;
        }

        out.println("<h2>Maximum Number is : "
        + max + "</h2>");
    }
}
```

Output -

Enter First Number : 10
Enter Second Number : 25

Maximum Number is : 25

Servlets can process form data sent using POST method and generate dynamic responses such as finding the maximum of two numbers.

Q1a) Explain the Following

i) Process of Transforming XML Document

XML transformation is the process of converting an XML document into another format such as: HTML, XHTML, PDF, Another XML document

This transformation is commonly done using **XSLT** (Extensible Stylesheet Language Transformations).

Need of XML Transformation

- To display XML data in browser
- To convert XML into user-friendly format
- To exchange data between systems
- To generate reports

Components Used

1. XML Document
2. XSLT Stylesheet
3. XSLT Processor

Transformation Process

Step 1: Create XML Document

Step 2: Create XSLT File

Advantages - Converts XML into readable format, Separates data and presentation, Reusable stylesheets

XML transformation using XSLT helps convert XML data into different formats for display and processing purposes.

ii) HTTP Session

An HTTP session is a mechanism used to maintain user-specific information across multiple requests in a web application.

HTTP protocol is stateless, so sessions are used to remember user data between requests.

Need of HTTP Session

- User login tracking
- Shopping cart management
- Online banking
- Personalized applications

HttpSession Object -

In Servlet, session management is done using HttpSession interface.

Methods of HttpSession -

Method	Purpose
getSession()	Creates or returns session
setAttribute()	Stores data in session
getAttribute()	Retrieves session data
invalidate()	Ends session

Example of HTTP Session

```
HttpSession session =
request.getSession();

session.setAttribute("user","Rahul");

String name = (String)session.getAttribute("user");
```

Advantages - Maintains user state, More secure than cookies, Stores data on server side

Disadvantages - Uses server memory, Session timeout issues

HTTP session helps web applications maintain user information and provide continuous interaction between client and server.

Q1a) Difference Between Server-Side Scripting Language and Client-Side Scripting Language

Basis	Server-Side Scripting	Client-Side Scripting
Definition	Script executed on the web server	Script executed on the client browser
Execution Place	Server	Client machine/browser
Purpose	Handles business logic, database operations	Handles user interface and validation
Processing Speed	Slower due to server communication	Faster because execution happens in browser

Basis	Server-Side Scripting	Client-Side Scripting
Security	More secure	Less secure
Database Access	Can access database directly	Cannot access database directly
Dependency	Depends on server	Depends on browser
Output	Sends processed HTML to browser	Manipulates webpage content dynamically
Visibility of Code	Hidden from user	Visible to user
Example Languages	PHP, JSP, Servlet, ASP.NET	JavaScript, VBScript
Internet Requirement	Requires server interaction	Can execute after page load
Main Use	Authentication, data processing	Form validation, animations